

PART 2

REQUIREMENT SPECIFICATIONS

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1 INTERPRETATION

- 1.1 The Interpretation Clause in the Conditions of Contract shall apply in the interpretation of this Requirement Specifications.

2 INTRODUCTION

- 2.1 This Part 2 (Requirement Specifications) sets out the framework for the establishment of the AI Delivery Partner Panel (the “Panel”), the scope of the Services to be provided, the evaluation process for appointment to the Panel, as well as the obligations of the Contractors in relation to the performance of Engagements.
- 2.2 Enablement, Transformations, Consulting (ETC) / GovTech Consulting (“GTC”) supports Public Sector Agencies in addressing complex digital and technology challenges, including the discovery, modernisation and delivery of digital systems and services.
- 2.3 The Authority intends to establish a time-bound Panel of contractors with demonstrated capability to analyse existing systems, establish mechanisms for verifying system behaviour and consistency, and develop working prototypes, replacement components or upgraded components using modern engineering approaches.
- 2.4 In this Requirement Specifications, “Panel” means the panel of successful Tenderers appointed, and “Panel Vendor” means a Tenderer appointed to the Panel; the Contractor. The Authority shall have the sole and absolute discretion to determine the number of Tenderers appointed to the Panel. The Authority reserves the right not to utilise any particular Panel Vendor during the Contract Period.
- 2.5 “Engagement” means a specific piece of work undertaken by a Contractor pursuant to a Request for Quotation (RFQ).
- 2.6 “Tech Lead” means the individual named by the Tenderer in its Tender Offer who shall personally participate in the live technical demonstration.
- 2.7 Engagements under the Panel are outcome-based assignments lasting approximately three (3) to nine (9) months. The actual scope, duration and value of each Engagement shall depend on the requirements identified by the Authority from time to time.
- 2.8 The Authority requires the Contractor to work hands-on with the Authority’s personnel and such other parties as may be designated by the Authority, and to

be capable of progressing from analysis of a current system and its actual behaviour through to working technical outputs.

- 2.9 The Services are not intended to constitute manpower augmentation. Contractors shall be expected to take responsibility for agreed technical outputs and outcomes rather than merely provide personnel on a time-and-material basis.
- 2.10 The Authority intends that Engagements be defined principally by the problem to be addressed, the outcomes to be achieved, applicable constraints and dependencies, success measures and required timeframe. The precise deliverables, acceptance criteria, commercial terms and other Engagement-specific requirements shall be set out in the relevant RFQ document.
- 2.11 Appointment to the Panel shall not constitute a representation, warranty or guarantee of any minimum number, volume or value of Engagements. Contractors appointed to the Panel to provide Services will be on an on-demand basis.

3 APPOINTMENT OF AI DELIVERY PARTNER PANEL

- 3.1 The Authority is inviting vendors with relevant expertise and experience to submit Tender Offers for evaluation to the Panel for the period in each Engagement.
- 3.2 The Authority's objective in establishing the Panel is to maintain a ready pool of contractors possessing sufficient technical capability, relevant personnel, delivery capacity and operating maturity to undertake the Services described the respective RFQ.
- 3.3 Appointment to the Panel shall be on an integrated-capability basis. Tenderers are expected to demonstrate credible capability across the core activities of current-state system analysis, consistency verification and specification-driven development, and prototype or modernisation delivery. The Authority shall be utilizing the Panel to meet its needs and requirements for AI modernisation and engineering services as and when it deems appropriate but will not confirm any volume of work or fees appointed to the Panel.
- 3.4 For the avoidance of doubt, the Authority reserves the right not to utilize any Panel Vendors. If in the Authority's view the appointed panel is unable to provide services for particular matters or types of matters (for example, if the Panel Vendors lack sufficient expertise in certain areas of AI modernization and engineering services). In addition, the Authority may, from time to time, issue fresh requests for proposals or tenders for services in respect of specific work or projects undertaken by the Authority.
- 3.5 For avoidance of doubt, the Authority shall have the sole and absolute discretion to determine the number of Panel Vendors who shall be appointed onto the Panel.

The appointment of a Panel Vendor to the Panel does not impose any obligation on the Authority to purchase any minimum quantity of Services from the Panel Vendor.

4 SCOPE OF SERVICES REQUIRED

- 4.1 The Authority is seeking Panel Vendors capable of providing the Services described in Appendix 1.
- 4.2 Where requested or directed by the Authority, the Services shall be provided at the premises of, and for the benefit of, any other Public Sector Agency.
- 4.3 The Services required comprise current-state system analysis, consistency verification, prototype and feasibility demonstration, incorporation of new or changed requirements, supporting documentation, tooling disclosure and specification-driven development.
- 4.4 Contractors may be required to perform any combination of the Services under a particular Engagement. The precise combination and scope shall be specified in the relevant RFQ.
- 4.5 The scope described in Appendix 1 is not exhaustive of every activity that may reasonably be required to achieve an Engagement outcome. A Contractor may be required to perform related technical, analytical or engineering activities reasonably incidental to the Services specified in the applicable RFQ.
- 4.6 Each Contractor shall be capable of working collaboratively with personnel of the Authority and such other agencies, contractors, users, product owners, technical specialists or other parties as may be designated by the Authority.
- 4.7 Where an Engagement involves software, prototypes, verification mechanisms, scripts, specifications, tests or other technical artefacts, the Contractor shall use appropriate engineering and quality practices and shall ensure that such outputs are capable of being understood, reviewed and, where applicable, continued or maintained by the Authority or another contractor.
- 4.8 Each Tenderer shall address its approach to specification-driven development in its methodology and technical approach as set out in Clause 5 below.
- 4.9 The Authority does not prescribe a particular software-development methodology, programming language, artificial intelligence product, automation product, specification language or proprietary technical approach unless otherwise stated for a particular Engagement in the RFQ.

- 4.10 The Contractor shall remain responsible for demonstrating that their methods and tools are suitable, secure, supportable and capable of achieving the outcomes specified by the Authority.

5 METHODOLOGY AND TECHNICAL APPROACH

- 5.1 Each Tenderer shall describe its methodology and technical approach for performing the Services, including the Tenderer's underlying technical reasoning, engineering judgement and decision-making process.
- 5.2 The Tenderer's methodology and technical approach shall explain how the Tenderer:
- a) understands and analyses the problem, existing system and available evidence;
 - b) develops, tests and refines hypotheses or technical conclusions;
 - c) identifies and manages uncertainties, assumptions, constraints and trade-offs;
 - d) makes and validates material technical and engineering decisions;
 - e) develops and maintains specifications or equivalent structured descriptions of required and existing behaviour;
 - f) progresses from analysis and specifications to implementation, prototype development or other technical outputs;
 - g) performs verification, validation and quality assurance; and
 - h) maintains appropriate traceability between requirements or established system behaviour, technical decisions, implementation or prototype development, and corresponding verification evidence.
- 5.3 The methodology and technical approach described by the Tenderer shall reflect its actual engineering practices and shall be substantiated by reference to one or more of the relevant track records and customer reference projects submitted as part of its Tender Offer. The Tenderer shall explain how the relevant methodology or practices were applied in the referenced project(s), including material technical decisions, challenges or constraints encountered, and how these were addressed.
- 5.4 The Authority does not require every element of the Tenderer's methodology to have been applied in a single reference project. Tenderers may rely on different

relevant track records to demonstrate the practical application of different aspects of their methodology and technical approach.

- 5.5 The Tenderer may sanitise, mask, redact or otherwise remove confidential, proprietary, personal, security-sensitive or customer-specific information from the methodology and technical approach proposal, provided that sufficient technical detail remains available for the Authority to assess the Tenderer's engineering practices, delivery workflow and use of methodology.
- 5.6 Generic or theoretical descriptions of methodology that are not supported by evidence of actual application may be accorded less weight in the evaluation.
- 5.7 The Tenderer shall submit its methodology and technical approach in accordance with the requirements set out in Part 3 (Guidelines for Tenderers). The methodology and technical approach will be evaluated in accordance with Part 4 (Evaluation Criteria).

6 VIDEO SUBMISSION AND TOOLING DISCLOSURE

- 6.1 As part of the tender evaluation, the Authority will evaluate the Tenderer's actual engineering practices, delivery workflow and use of relevant tooling, including, where applicable, automation and artificial intelligence-enabled tools, through the Tenderer's video submission and tooling disclosure.
- 6.2 The Tenderer shall provide its tooling disclosure and video submission as part of its Tender Offer in accordance with Annex IV of Part 3 (Guidelines for Tenderers).
- 6.3 The tools shall identify the relevant tools, platforms and/or services used by the Tenderer in performing services of the nature described in this Requirement Specifications. Where automated or artificial intelligence-enabled tooling is used, the Tenderer shall describe:
- a) the function performed by the tooling;
 - b) the material inputs to and outputs from the tooling;
 - c) the parts of the engineering workflow that remain dependent on human analysis, engineering judgement, review or intervention; and
 - d) the controls used to review, validate or otherwise assure the outputs produced by the tooling.
- 6.4 The video submission shall demonstrate the Tenderer's actual engineering practices and delivery workflow in operation and shall provide sufficient technical

detail for the Authority to understand how the Tenderer applies its disclosed tooling and engineering practices in practice. For the avoidance of doubt, the Authority does not require the Tenderer to have developed or own any proprietary tool or technology. Tenderers may demonstrate the use of proprietary, commercially available, open-source or other third-party tools and services. The evaluation will focus on how the Tenderer applies the relevant tools within its engineering workflow, including the Tenderer's underlying approach, engineering judgement, human review and controls, rather than on ownership of the tools themselves.

6.5 The video submission shall demonstrate:

- a) how the Tenderer progresses from understanding an existing system, requirement or other technical input to implementation, prototype development or other technical output;
- b) how specifications or equivalent structured descriptions of required or existing behaviour are developed, maintained and used within the engineering workflow;
- c) how relevant tooling, including automation or artificial intelligence-enabled tooling, is incorporated into that workflow;
- d) how human analysis, engineering judgement, review and validation are applied; and
- e) how implementation or prototype outputs are verified and how relevant traceability or verification evidence is maintained.

6.6 The video submission may use an existing or completed project, or a representative environment or example that reasonably demonstrates the Tenderer's actual engineering practices. The Tenderer is not required to develop a new solution solely for the purpose of preparing the video submission.

6.7 The Tenderer may sanitise, mask, redact or otherwise remove confidential, proprietary, personal, security-sensitive or customer-specific information from the video submission, provided that sufficient technical detail remains available for the Authority to assess the Tenderer's engineering practices, delivery workflow and use of tooling.

6.8 The video submission and tooling disclosure will be evaluated in accordance with the applicable evaluation criterion in Part 4 (Evaluation Criteria). For the avoidance of doubt, the Video Submission is separate from, and does not replace the live technical demonstration.

6.9

7 LIVE TECHNICAL DEMONSTRATION

- 7.1 The Authority may invite Tenderers to participate in a live technical demonstration for evaluation purposes.
- 7.2 A Tenderer invited to participate in the live technical demonstration shall demonstrate its actual technical approach, engineering workflow and relevant tooling for the Services described in this Requirement Specifications, with particular emphasis on specification-driven development.
- 7.3 The demonstration shall show how the Tenderer progresses from understanding an existing system or requirement through to the creation or refinement of specifications, implementation or prototype development, and verification of the resulting behaviour or output.
- 7.4 Where automated or AI-enabled tooling is demonstrated, the Tenderer shall explain the specific function performed by the tooling, the material inputs and outputs, the parts of the workflow that remain dependent on human analysis or engineering judgement, and the controls used to review or validate the resulting output.
- 7.5 The demonstration may be based on an existing or completed implementation delivered for a public sector agency, private sector client, or a representative environment that reasonably demonstrates the relevant capability. The Tenderer is not required to develop a new solution solely for the purpose of the Tender.
- 7.6 The Tenderer may sanitise, mask, redact or otherwise remove confidential, proprietary, personal, security-sensitive or customer-specific information from the material demonstrated, provided that sufficient technical detail and functionality remain available for the Authority to assess the Tenderer's actual approach and capability.
- 7.7 The Tech Lead named in the Tender Offer shall personally participate in and lead the live technical demonstration. The demonstration shall focus principally on technical work and actual engineering practice. Other personnel may attend and support the demonstration where appropriate, but the Authority may direct technical questions to the Tech Lead personally.
- 7.8 The Authority may ask questions during or following the demonstration to clarify the Tenderer's technical approach, the role of the Tech Lead, the tooling used, the degree of automation, technical decisions made, limitations encountered and the relevance of the demonstrated work to the Services.

- 7.9 The live technical demonstration shall comprise approximately 20 minutes for the demonstration and 10 minutes for questions and answers, unless otherwise notified by the Authority before the demonstration.
- 7.10 The Authority will notify invited Tenderers of the date, format, access arrangements and any administrative instructions for the live technical demonstration.
- 7.11 The Tenderer shall remain responsible for demonstrating that their methods and tools are suitable, secure, supportable and capable of achieving the outcomes specified by the Authority.

8 REQUEST FOR QUOTATION

- 8.1 All Purchase Order(s) shall be issued by the Authority through the Enablement, Transformations Consulting (ETC) team / GovTech Consulting (GTC), or any person duly authorised by GTC.
- 8.2 The Authority may issue an RFQ to any one or more Panel Vendors at any time and from time to time during the Contract Period.
- 8.3 A Panel Vendor issued with an RFQ shall submit its quotation within the period and in the manner specified by the Authority.
- 8.4 Each RFQ may specify the problem or opportunity to be addressed, desired outcome, constraints, dependencies, available artefacts, success measures, required deliverables, acceptance criteria, timeframe, required personnel or capabilities, security requirements and commercial terms. Each Panel Vendor shall respond to the RFQ on an outcome-based basis and shall explain how its proposed approach, team, delivery plan and commercial model are aligned to the relevant outcome and success measures.
- 8.5 The Authority may require a Panel Vendor to identify assumptions, dependencies, risks, matters within the Authority's control and matters within the Panel Vendor's control that may materially affect achievement of the specified outcome.
- 8.6 Commercial terms applicable to an Engagement may include payment linked to agreed deliverables, milestones, outcomes, success measures or other performance conditions.
- 8.7 A Panel Vendor shall commence an Engagement only after the Authority issues the Purchase Order in response to the Panel Vendor's response to the RFQ.

- 8.8 The Contractor shall provide the Services in accordance with the Purchase Order Contract within the period stipulated therein.
- 8.9 Nothing in the Master Contract shall be construed as (whether expressly or impliedly):
- a) Requiring the Authority to issue any Purchase Order to any Panel Vendors;
 - b) Granting any Panel Vendor any exclusive right to provide Services to the Authority;
 - c) Preventing or restricting the Authority from obtaining Services from other Panel Vendors, whether that Panel Vendor is part of the Panel or otherwise;
 - d) Preventing or restricting the Authority from calling any fresh quotation and/or tender to procure Services during the term of the Master Contract.

9 KEY PERSONNEL AND DELIVERY CAPACITY

- 9.1 The Contractor shall propose Key Personnel for each engagement as part of its quotation in response to the relevant RFQ. The proposed Key Personnel shall collectively demonstrate credible capability across the Services required under the RFQ and shall include one individual designated as the Tech Lead.
- 9.2 The Authority does not prescribe fixed job titles for the Key Personnel. The Contractor may propose the combination of technical roles they consider most appropriate, provided that the proposed personnel collectively demonstrate the relevant capabilities to provide the Services stated in the RFQ.
- 9.3 The Contractor shall complete the Key Personnel Declaration Form set out in Annex I of Part 2 for each Key Personnel proposed, as part of its quotation submitted in response to the relevant RFQ. Each Key Personnel shall provide the information required in the Key Personnel Declaration Form, including relevant experience across the Services, where applicable.
- 9.4 The Contractor shall confirm the availability of the Key Personnel proposed for that Engagement and identify any other personnel proposed for deployment.
- 9.5 The Contractor shall not substitute the Tech Lead or any other Key Personnel without informing the Authority.
- 9.6 Any proposed replacement (including the Tech Lead) shall possess capability, relevant experience and seniority equivalent to or greater than the individual being replaced.

- 9.7 The Contractor shall provide the proposed replacement's relevant project information, references, availability, security-clearance information and such other information as the Authority may require. The Authority may interview, technically assess or otherwise verify the capability of any proposed replacement.
- 9.8 The Contractor shall ensure adequate handover between outgoing and replacement personnel. Any overlap or handover arising from a Contractor-initiated substitution shall be provided at no additional cost to the Authority, unless otherwise agreed in writing.
- 9.9 The Contractor shall not be entitled to rely on a delay in obtaining security clearance for a replacement as a reason for delayed performance where the replacement resulted from circumstances within the Contractor's reasonable control.
- 9.10 Where security-clearance delay arises solely from Authority processing after the Contractor has provided all requested information accurately and promptly, the parties shall address any resulting schedule impact.
- 9.11 The Contractor shall demonstrate sufficient bench strength beyond the Key Personnel proposed for an Engagement to support continuity and, where required, concurrent Engagements.
- 9.12 The Authority intends, where practicable, to initiate security clearance for the successful Tenderer's named Key Personnel promptly after Panel appointment so that suitable personnel may be available for early deployment. The Contractor shall provide promptly all information, declarations and assistance reasonably required for such clearance.
- 9.13 The Authority may, from time to time and as it considers necessary, require the Panel Vendor to attend review meetings with the Authority to discuss matters relating to the Panel Vendor's delivery capacity, bench strength, availability of relevant personnel, status and performance of ongoing Engagements, and such other matters relevant to the Panel Vendor's continued ability to perform the Services. The Panel Vendor shall provide such information and updates as the Authority may require for the purposes of such review.

10 QUALITIES AND STANDARDS

- 10.1 Each Contractor and its Key Personnel shall exercise all reasonable care, skill and diligence in accordance with good and sound professional, technical and engineering practice.
- 10.2 Each Contractor shall ensure that the requirements set out in the Master Contract and applicable Purchase Order Contract are met.
- 10.3 Each Panel Vendor shall ensure consistent and adequate technical supervision of junior or less experienced personnel by personnel with sufficient seniority, competence and experience.
- 10.4 Each Contractor shall perform the Services with reasonable dispatch, having regard to the timeframes stipulated by the Authority. Time shall be of the essence of the Purchase Order Contract.
- 10.5 The Contractor shall inform the Authority as early as reasonably practicable of any matter likely to delay or materially affect achievement of an Engagement outcome and shall propose appropriate corrective or mitigating action.
- 10.6 Where software or technical artefacts are produced, the Contractor shall apply appropriate quality practices, including such automated testing, code review, static or dynamic analysis, secure coding, continuous integration, traceability, peer review or other practices as are appropriate to the Engagement.
- 10.7 The exact quality standards, policies, procedures and technical controls applicable to an Engagement may be specified by GTC in the relevant RFQ.

11 PRICE SCHEDULE AND COMMERCIAL TERMS

- 11.1 No price schedule is required as part of the Tender Offer for appointment to the Panel. The Authority will not be evaluating price as part of the Tender evaluation for appointment to the Panel.
- 11.2 Pricing and commercial terms applicable to each Engagement shall be determined separately through the applicable RFQ process. An Engagement may include commercial provisions under which part of the amount payable is dependent on achievement of specified deliverables, outcomes, success measures, delivery timing or other performance conditions
- 11.3 Where GTC issues an RFQ, the Panel Vendor shall submit the pricing information required in the format specified by the Authority. The specific payment

mechanism, including any amount linked to outcome or performance, shall be stated in the relevant RFQ.

11.4 The Authority may require pricing to be expressed as a fixed amount, milestone-based amount, outcome-based amount, capped amount or such other commercial basis as it considers appropriate.

11.5 The Panel Vendor shall ensure that any commercial proposal states all material assumptions, dependencies, exclusions and conditions affecting the price.

12 PERSONNEL SECURITY CLEARANCE

12.1 The Panel Vendor shall ensure that its personnel, including personnel of any approved Subcontractor, obtain such security clearance as may be required before being given access to relevant information, systems, environments or premises.

12.2 Each Panel Vendor individual shall only be permitted access to information of a classification commensurate with that individual's approved clearance. Access shall be granted on a need-to-know and least-privilege basis and only to the extent required for performance of the individual's responsibilities.

12.3 The Panel Vendor shall define and communicate the roles and responsibilities of all personnel involved in an Engagement.

12.4 The Panel Vendor and its personnel shall comply with all applicable security requirements and written security directions notified by the Authority for the relevant Engagement, including any requirements derived from applicable Government ICT and security policies.

12.5 The Authority may reject or require removal of any person who fails to obtain or maintain required clearance or whom the Authority reasonably considers unsuitable for security, access or operational reasons.

12.6 Where known and appropriate for disclosure, GTC will state relevant security, clearance and access requirements in the applicable RFQ.

12.7 Any indicative clearance timeframe communicated by the Authority shall be for planning purposes only and shall not constitute a guarantee that clearance will be completed within that timeframe.

13 INFORMATION HANDLING

- 13.1 The Panel Vendor shall not transfer security-classified information, sensitive information or personal data obtained or held in connection with the Master Contract or the Purchase Order Contract outside Singapore, or allow any party outside Singapore to access such information or data, without the Authority's prior written approval.
- 13.2 The Panel Vendor shall not disclose information, including its source, received or generated in connection with the Master Contract or any Purchase Order Contract except where specifically authorised by the Authority in writing or otherwise permitted under the Master Contract or Purchase Order Contract.
- 13.3 The Panel Vendor shall use information received in connection with an Engagement solely for the purpose of performing that Engagement unless otherwise approved by the Authority.
- 13.4 The Panel Vendor shall be responsible for safeguarding information made available to it and shall ensure that its personnel, agents and approved Subcontractors safeguard such information.
- 13.5 The Panel Vendor shall maintain appropriate administrative, technical, physical and personnel controls to protect information against accidental or unlawful loss and against unauthorised access, disclosure, copying, use, alteration, deletion or modification.
- 13.6 Where required, the Panel Vendor shall provide a detailed description of the systems, measures, technologies, processes and procedures used to manage and protect information.
- 13.7 The Panel Vendor shall ensure that personnel do not remove or retain Authority information upon leaving an Engagement or when such information is no longer required.

14 DATA CONFIDENTIALITY AND INTEGRITY

- 14.1 The Contractor shall implement appropriate controls to protect the confidentiality and integrity of credentials, source code, system information, data, test data, specifications, documentation and other information.
- 14.2 The Contractor shall implement measures to prevent unauthorised access to, disclosure of, modification of or deletion of Authority information.
- 14.3 The Contractor shall implement appropriate controls to protect the confidentiality and integrity of credentials, source code, system information, data, test data, specifications, documentation and other information.
- 14.4 The Contractor shall notify the Authority promptly in accordance with the Master Contract or the Purchase Order Contract, if it becomes aware of any actual or suspected unauthorised disclosure, access, loss, alteration or compromise.
- 14.5 The obligations under this Clause shall survive expiry or termination of the Master Contract and any Purchase Order Contract.

APPENDIX 1: DETAILED SCOPE OF SERVICES**1 CURRENT STATE SYSTEM ANALYSIS**

- 1.1 The purpose of the current-state system analysis is to identify and document the system's actual logic, behaviour, business rules, dependencies and data flows.
- 1.2 The Contractor may be required to directly examine an existing system's source code together with available business requirements, business cases, requirement specifications, operating information, architecture information and other relevant documentation.
- 1.3 The Contractor shall not assume that existing documentation accurately or completely represents the current system.
- 1.4 Where source code is available, the Contractor may be required to establish actual system behaviour through direct code inspection supplemented by configuration, data structures, interfaces, tests, runtime evidence and such other sources as are appropriate.
- 1.5 Where source code is unavailable, inaccessible, incomplete or unsuitable for reliable analysis, the Contractor shall use other available evidence to establish relevant system behaviour. Such evidence may include controlled test inputs and outputs, application interfaces or APIs, operational logs, historical transactions, database observations, configuration information, test artefacts, user workflows and runtime observation.
- 1.6 The Panel Vendor shall document the basis on which it has inferred existing behaviour and shall identify material assumptions, evidence limitations and areas of uncertainty.
- 1.7 The Panel Vendor shall identify material discrepancies, ambiguities or gaps between available documentation and observed or established system behaviour where relevant to the Engagement.

2 CONSISTENCY VERIFICATION

- 2.1 The Panel Vendor may be required to design and build a mechanism that captures relevant behaviour of an existing system, including actual inputs, outputs, business rules or other observable behaviour.
- 2.2 The mechanism shall enable the Authority to determine whether a newly developed replacement or upgraded component produces behaviour consistent with the existing system where equivalence is intended.

- 2.3 Where the Authority has deliberately approved a change to existing behaviour, the mechanism shall identify the expected difference and verify the replacement or upgraded component against the approved changed requirement.
- 2.4 The mechanism shall be repeatable to the extent reasonably practicable and shall provide sufficient evidence for the Authority to understand what was verified, against which expected behaviour or specification, and with what result.
- 2.5 Where the behavioural baseline is derived from black-box observation rather than source-code analysis, the Panel Vendor shall explain the evidence used, coverage achieved, known limitations and level of confidence reasonably attributable to the resulting verification.
- 2.6 The Panel Vendor shall distinguish any part of the consistency verification process that relies on manual judgement from any part that is automated.

3 PROTOTYPE AND FEASIBILITY DEMONSTRATION

- 3.1 Using documented current-state logic and applicable specifications as a basis, the Panel Vendor may be required to design and build working prototypes of a replacement system, upgraded system or selected component.
- 3.2 Where the Authority requires a Proof of Concept, the prototype shall demonstrate technical feasibility against criteria specified in the relevant Engagement.
- 3.3 Where the Authority requires a Proof of Value, the prototype shall demonstrate measurable benefit against the baseline or success measure specified by the Authority.
- 3.4 The required level of fidelity shall be stated or agreed for each Engagement. Unless expressly required, a prototype need not constitute production-ready software.
- 3.5 Conclusions concerning feasibility, limitations, risks and recommended next steps shall be supported by evidence generated during the Engagement.

4 INCORPORATING NEW OR CHANGED REQUIREMENTS

- 4.1 The Panel Vendor's approach shall enable new business rules, policy changes, technical constraints or other requirements identified during an Engagement to be incorporated into the analysis, specifications, verification mechanism and prototype or replacement component.

- 4.2 The Panel Vendor shall manage such changes in a manner that preserves reasonable traceability to prior analysis and does not require the analysis to restart unless materially necessary.
- 4.3 The Panel Vendor shall identify the impact of a changed requirement on existing specifications, expected behaviour, verification evidence and technical outputs.

5 SUPPORTING DOCUMENTATION

- 5.1 The Panel Vendor may be required to produce and maintain structured supporting artefacts including requirement specifications, business requirement documents, technical notes, design references, comparison matrices, test or verification artefacts and decision records.
- 5.2 Such artefacts shall be maintained as documented inputs to keep the current-state analysis, specifications and consistency verification mechanism accurate and sufficiently complete as the Engagement progresses.
- 5.3 Documentation shall be proportionate to the needs of the Engagement and capable of being reasonably understood, used and continued by the Authority or another contractor.
- 5.4 Unless otherwise agreed, final documentation shall be provided in an editable form.

6 TOOLING DISCLOSURE

- 6.1 Where automated tooling is used to accelerate any part of the Services, including source-code analysis, document analysis, requirements analysis, code generation, test generation, consistency verification or prototype development, the Contractor shall disclose the relevant tooling.
- 6.2 The Contractor shall identify what the tooling performs or automates, its material inputs and outputs and those parts of the work that remain dependent on manual analysis, judgement, review or engineering.
- 6.3 Where artificial intelligence is used, the Contractor shall explain the specific function performed by the tool and the human controls applied to its output.
- 6.4 Use of any tool within an Authority-controlled environment or use of Authority information as input to any external tool or service shall be subject to the Authority's prior approval and applicable security requirements.

7 SPECIFICATION-DRIVEN DEVELOPMENT

- 7.1 The Contractor shall use an approach under which specifications, or equivalent structured descriptions of required and existing behaviour, serve as an active input to both implementation and verification.
- 7.2 The Contractor shall establish and maintain reasonable traceability between relevant current-state findings, approved new or changed requirements, the applicable specification, implementation or prototype and corresponding verification evidence.
- 7.3 The Contractor shall explain how specifications are created, reviewed, versioned and updated as understanding of the current system or required future behaviour evolves.
- 7.4 The Contractor shall explain how the specification informs the build process and how the same specification, or a related authoritative representation, informs consistency verification.
- 7.5 The Authority does not prescribe a particular specification language, modelling notation, test framework or technical tool.

8 TOOLS, LICENCES, AND USAGE CHARGES

- 8.1 Unless otherwise specified in the RFQ, the Authority will provide the computing environment and access to such Authority-managed development and collaboration tools as are required for the Engagement and made available by the Authority. These may include SEED devices and access to tools such as GitLab, Jira, Confluence and SonarQube.
- 8.2 The Authority may also provide access to approved application programming interfaces, credentials, token usage, API keys or other Authority-managed technical services where required for the Engagement. The availability and permitted use of such services shall be subject to the Authority's applicable security, usage and technical requirements.
- 8.3 The Contractor shall ordinarily provide, as part of its quoted price and at its own cost, the standard tools, software, subscriptions, licences and other resources forming part of its normal delivery capability, unless the RFQ expressly states that such items will be provided or separately paid for by the Authority.
- 8.4 Except for any infrastructure, tools, equipment or technical services provided, or stated by the Authority to be provided, pursuant to Clauses 8.1 and 8.2 or the relevant RFQ, the Contractor shall, at no additional cost to the Authority, provide its personnel with all infrastructure, tools and equipment necessary for the

performance of the Services for the Engagement. Such infrastructure, tools and equipment shall include, where applicable, notebooks, coding and development tools, productivity tools such as Microsoft Office, integrated development environments such as Eclipse, Visual Studio and RubyMine, and other standard tools required for the Contractor's normal delivery capability. The Contractor shall ensure that any Contractor-provided devices, computing infrastructure and equipment used for the Engagement have appropriate anti-virus or anti-malware protection software installed and that such software, including the applicable definitions or signatures, is kept up to date.

- 8.5 The Contractor shall ensure that any tool, software, platform or service provided or selected by the Contractor for use in the performance of the Services is maintained at the latest generally available stable and supported version, release or update made available by the relevant provider, subject to applicable security, technical, compatibility, licensing and approval requirements. Where a later version, release or update becomes available during an Engagement, the Contractor shall implement such version, release or update as soon as reasonably practicable, unless otherwise approved in writing by the Authority.
- 8.6 Where the Contractor considers that an additional third-party specialist tool, software licence, API, cloud service or other usage-based service not provided by the Authority is necessary or materially beneficial for an Engagement, the Contractor shall identify the proposed item separately in its RFQ response.
- 8.7 The Contractor shall state the purpose of the proposed item, the applicable licensing or charging basis, estimated quantity or usage, period of use, assumptions affecting the charge, and any material licensing, hosting, data-location or usage restrictions relevant to the Authority.
- 8.8 The Contractor shall not incur any separately chargeable licence, subscription, API, consumption or similar third-party cost on behalf of the Authority unless that cost has been approved in writing by the Authority and incorporated into the applicable Purchase Order Contract.
- 8.9 Unless otherwise expressly agreed, any approved third-party charge reimbursed by the Authority shall be limited to the actual approved cost incurred by the Contractor and shall be supported by reasonable documentary evidence, including applicable invoices or usage records. No mark-up shall apply unless expressly approved by the Authority.
- 8.10 The Authority shall not be obliged to approve any tool, licence or third-party service proposed by the Contractor. Where approval is not granted, the Contractor shall, where reasonably practicable, propose an alternative approach that does not materially compromise the required Engagement outcome.

- 8.11 All tools and third-party services used in connection with Authority information or Authority-controlled environments shall remain subject to the security and tooling requirements in this Requirement Specifications and the relevant Purchase Order Contract.

ANNEX I: KEY PERSONNEL DECLARATION FORM

I PERSONAL PARTICULARS

Name of Staff : _____
Designation : _____
Tech Lead (Yes / No) : _____
Principal capability area(s) : _____
Citizenship¹ : _____
Country of Residence : _____
Proposed Role in the Services : _____
Years of relevant experience: _____

II PROFILE (Brief description of past and present work portfolio)

III EDUCATION / PROFESSIONAL QUALIFICATIONS / CERTIFICATIONS

<u>Period</u>	<u>Discipline / University (Name and Country) /</u>
<u>Certifications</u>	

IV EMPLOYMENT HISTORY

<u>Period</u>	<u>Appointment / Organisation</u>	<u>Responsibilities</u>
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V PROFESSIONAL / TECHNICAL EXPERIENCE RELEVANT TO THE SERVICES

(e.g. experience in current-state system analysis, software or system understanding, consistency or behavioural verification, specification-driven development, software or prototype development, testing, automation and code-quality practices; include length and period of relevant exposure.)

¹ Please indicate whether “Permanent Resident” or “Employment Pass Holder” if non-Singaporean.